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Wide-Area Acoustic Monitoring for Subsea Integrity Assurance: Field Validation of a Broadband Active Sonar System

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Abstract

Hydrocarbon and CO_2 leakage from subsea infrastructure presents a significant risk to offshore safety and environmental integrity. This paper describes the development, deployment, and field validation of a wide-area acoustic monitoring system for subsea leak detection and integrity assurance. Originally developed by Metas AS and now owned by Imenco Future Technologies AS, the system employs broadband sonar and machine learning algorithms to detect and classify acoustic anomalies such as gas plumes and oil droplets. The modular platform supports both standalone operation and integration with offshore Safety and Automation Systems (SAS) in accordance with API 17F. Field deployments, including multi-year monitoring at an Equinor production site, demonstrated detection ranges exceeding 1,000 meters for gas-phase releases and four hundred meters for denser fluids, with low false alarm rates and stable classification under varying environmental conditions. These capabilities enable persistent leak monitoring and early anomaly detection. The system architecture allows operators to tailor coverage and integrate alerting functions into operational systems. This field-proven acoustic monitoring solution supports proactive risk management and complements surface monitoring tools such as satellite-based SAR, especially in detecting and localizing subsea leak sources.

Introduction, Problem Statement, and State of the Art

Hydrocarbon and CO_2 leakage from subsea infrastructure remains one of the most challenging integrity and environmental risks in offshore operations (Norwegian Petroleum Safety Authority, 2024). Such leaks can result in regulatory non-compliance, environmental damage, costly remediation, and reputational harm (OGP, 2013). Offshore oil and gas operators are under increasing scrutiny from regulators, stakeholders, and the public to detect, localize, and mitigate leaks quickly and reliably, particularly as subsea infrastructure ages.

Limitations of Traditional and Surface-Based Detection

Traditional inspection methods—such as ROV or diver visual inspection, periodic pressure testing, and ad hoc observation—are effective for confirming known defects but are episodic in nature (DNV, 2021). They cannot provide continuous coverage or reliably detect small, intermittent leaks before they escalate.

Surface-based detection technologies include:

- Visual observation from platforms, vessels, or aircraft (per Guideline 100): low cost, but limited to daylight, good weather, and larger spills (Offshore Norge, 2023).
- Radar-based systems such as Oil Spill Detection (OSD) processors on X-band navigation radar: capable of continuous monitoring in certain wind ranges (typically 5–12 m/s) but only detect oil on the surface and cannot localize subsea sources (Johannessen et al., 2006).
- Thermal and optical imaging (EO/IR cameras): Infrared and optical imaging techniques can confirm and map slicks on the sea surface, but their effectiveness depends on external factors such as oil thickness, ambient light, and viewing geometry (Fingas and Brown, 2018).
- Satellite SAR: widely used for regional surveillance and enforcement but cannot detect small leaks or operate outside moderate wind conditions (ScanEx, 2018). It also provides no direct link to the subsea source location.

These approaches are valuable for surface slick detection, as seen in Figure 1, and compliance monitoring, but they cannot reliably identify low-volume subsea releases before they breach the surface, especially for heavier oils or CO_2 leaks that may dissolve or disperse before reaching it (Offshore Norge, 2023; Prakoso et al., 2022).

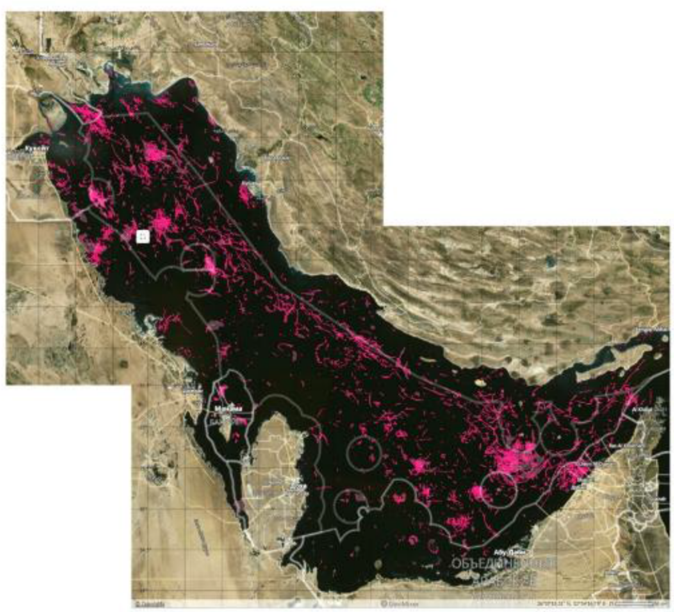

Figure 1—Summary Map of all oil pills detected in the Persian Gulf in 2017 from Satellite based SAR
Source: https://www.scanex.ru/en/company/news/satellite-monitoring-of-oil-pollution-in-the-persian-gulf/

Subsea Leak Detection Methods

Below the surface, detection methods include:

- Passive acoustics: Detects sound generated by leak turbulence, but performance is highly dependent on conditions, fluids types (oil, water, gas) released and fractions, leak rate, differential pressure, and background noise (Prakoso et al., 2022). Signal attenuation in seawater and masking

by ambient noise often limit practical range, especially for liquid-phase leaks and distributed assets such as pipelines.
- Chemical sensing: Detects dissolved hydrocarbons in water samples or via in-situ probes. While highly sensitive, these systems typically have small spatial coverage and require proximity to the leak (Reddy et al., 2012).
- Optical techniques: Laser fluorosensors HLIF LiDAR offers direct hydrocarbon detection in the water column, though its application is restricted to relatively short ranges (tens to hundreds of meters) and it performs best in clear water conditions (Jha et al., 2008)

Each method presents trade-offs between sensitivity, range, persistence, and applicability to different leak types. No single technology provides comprehensive coverage in all scenarios, making complementary systems essential (Offshore Norge, 2023).

The Role of Wide-Area Active Sonar

Wide-area active sonar bridges many of the gaps in existing monitoring strategies. By emitting broadband acoustic pulses and analyzing the backscatter, active sonar can:

- Detect both gas and liquid-phase plumes in the water column before they reach the surface (Kublius, 2015; Loranger, 2019).
- Operate independently of visibility, daylight, and most weather conditions (Pedersen and Kublius, 2019).
- Provide volumetric coverage on the order of 1–2 km radius from a single deployment (Metas AS, 2022).
- Classify detected targets using frequency response and spatial dynamics to distinguish leaks from biological or man-made objects (Kublius, 2015).

The WAAM system, originally developed by Metas AS and now part of Imenco Future Technologies AS through acquisition, builds on over a decade of active acoustic research and subsea leak detection trials (Kublius, 2015; Loranger, 2019, Pedersen and Kublius, 2019). Unlike surface-based tools, WAAM is a seabed-deployed, self-contained sensor package that continuously scans the full water column around critical assets. Its programmable scanning regimes, wideband transducer compatibility, and onboard machine learning classification enable persistent, automated detection with low false alarm rates, one in five years of operation (Metas AS, 2022).

Need for Integrated Detection Strategies

Guideline 100 emphasizes the selection of Best Available Techniques (BAT) based on the specific environmental risk, fluid type, and asset configuration (Offshore Norge, 2023). No single method is universally optimal; combining technologies is often the most robust strategy. For example:

- SAR or aerial surveillance for regional spill mapping (ScanEx, 2018).
- OSD radar for continuous platform vicinity monitoring Johannessen et al., 2006).
- WAAM or similar active sonar systems for subsea leak localization (Kublius, 2015).
- Chemical or optical sensors for corroboration and quantification (Reddy et al., 2012; Jha et al. 2008).

In this context, WAAM offers a unique capability: continuous, wide-area subsea monitoring that complements surface detection systems, fills the early-detection gap, and directly addresses Norwegian Activities Regulation §57's requirement for timely leak detection and mapping independent of environmental conditions (Norwegian Petroleum Safety Authority, 2024).

Technology Overview

The WAAM system (Figure 2) is built around a compact, seabed-mounted sonar platform with a highly adaptable scanning and data processing architecture (Kublius, 2015). Beam widths for commonly used transducers typically range around 7°, but broader or narrower configurations are supported depending on operational requirements (Kongsberg Discovery, 2024). The programmable scanning regime allows full 360° horizontal coverage and -10° to +60° vertical range or can be narrowed to specific arcs and sectors for efficiency (Metas AS, 2022). Each scan is composed of horizontal segments at different vertical tilt steps, with ping overlap configurable to optimize spatial resolution. An example could be 90° horizontal scan with five vertical steps which could include seventy-five pings and complete a full volume scan in 80–90 seconds with a one degree overlap in pings (Metas AS, 2022).

Figure 2—Image of WAAM Sensor Source: Imenco Future Technologies

Scan frequency, angular resolution, and data refresh rates are tunable to match application-specific demands, such as leak detection, biomass monitoring, or anomaly tracking (Kublius, 2015). The broadband active sonar operates in wide frequency bands to capture the full spectral signature of underwater targets, supporting classification based on frequency response and target dynamics (Pedersen and Kublius, 2019).

On the processing side, WAAM employs a modular data pipeline architecture that allows both serial and parallel evaluation of incoming sonar data. The onboard software supports real-time filtering, aggregation, and conditional analysis of classification outputs (Imenco Software Release Notes, 2023). Detected targets are initially sorted into one of three categories: man-made objects, biologics, or leak/seep events (Kublius, 2015). For leak detections, a secondary pipeline calculates location, estimates volumetric release rate, assesses plume stability and trend, and infers likely fluid type based on acoustic behavior (Imenco Software Release Notes, 2023).

The classification engine is based on supervised machine trained models, originally trained on labeled datasets from laboratory and field conditions (Kublius, 2015). Model retraining can be done periodically using field-verified detection examples, and system parameters are updateable via remote firmware packages (Imenco Software Release Notes, 2023). All processed outputs are available via SIIS Level 3 and Modbus interfaces, enabling direct integration with SAS (Safety Automation System) or DCS

(Distributed Control System) architecture over OPC UA (API, 2017). The system supports autonomous and event-triggered scanning modes, providing operators with flexible deployment strategies across subsea production, CCS, or decommissioning contexts.

The WAAM unit is a self-contained acoustic monitoring system measuring approximately 44 cm in diameter and 120 cm in height. The system is typically deployed within a lander structure, integrated on existing subsea equipment (Figure 3), or embedded in prefabricated concrete mattresses for stable seabed deployment (Metas AS, 2022).

Figure 3—ROV Screenshot of WAAM installed on Z2 structure at Well Site H Source: Deepocean courtesy of Equinor.

WAAM supports the full suite of Kongsberg Discovery's wideband scientific echosounders, typically operating with one transducer at a time. While common configurations include the Simrad ES70 and ES333, with center frequencies of 70 kHz for longer range monitoring and 333 kHz for smaller areas, respectively, the system is also compatible with a broader range of wideband variants offering beam widths optimized for acoustic localization, allowing tailored deployments for different detection scenarios (Kongsberg Discovery, 2024). This broader capability allows for system optimization across a wide frequency range depending on target range, fluid type, and background noise. The sonar operates in broadband mode, capturing both signal strength and frequency-dependent target responses across a spectrum to aid in classification (Loranger, 2019, Pedersen and Kublius, 2019).

The scanning regime is fully programmable and can be tailored to the specific geometry and monitoring needs of a given field. Scans are conducted as horizontal segments at each vertical step, which are then combined into a full volume scan. For example, a scan covering a 90° horizontal arc with five vertical steps, each with 1° horizontal ping overlap, would result in fifteen pings per vertical layer, totaling approximately seventy-five pings. With a typical ping rate of 1 Hz, a complete volume scan would be completed approximately every 80–90 seconds, depending on overlap and tilt transition time (Metas AS, 2022). By adjusting scan geometry—such as reducing horizontal span or vertical coverage—operators can decrease scan cycle time and increase temporal resolution.

Similarly, scan resolution can be increased by slowing the scan speed, resulting in greater overlap between pings and enhanced localization precision. Vertical scan coverage is typically adjusted based on water depth, with shallow environments requiring less angular range to reach the surface (Kublius, 2015). This flexibility enables operational optimization across a wide range of deployments.

The WAAM system uses an adaptable software architecture that supports multiple data pipelines processed both in series and in parallel. This modular framework allows for the application of data filters, cross-validation routines, and targeted analytics based on the type of target identified (Metas AS, 2022). For leak detection, once a target is categorized as a leak or seep, a dedicated analysis pipeline is activated. This includes extraction of target location, estimation of release rate, trend analysis of flow consistency across scans, and fluid-type characterization (Metas AS, 2022). These functions are critical for supporting both alarm confirmation and operational decision-making.

The acoustic data stream is processed locally on embedded computing hardware using supervised machine learning classification algorithms (Kublius, 2015). These models were initially trained on labeled field data from known targets and validated across various leak scenarios. All detected targets are currently classified into one of three categories: (1) man-made objects (e.g., risers, anchors), (2) biologics (fish, marine mammals, etc.), or (3) leak/seep sources. Classification confidence and trend tracking are included in the telemetry.

Processed outputs are transmitted through Modbus or SIIS Level 3 compliant interfaces. The WAAM system's compact design, low power consumption (<35W at 24VDC), and modular electronics, capable of remote firmware updates, enable flexible integration in brownfield environments (Imenco Product Datasheet, 2024). The system supports both autonomous and externally triggered scan modes, allowing use in persistent surveillance or short-duration integrity testing.

Methodology

Field validation of the WAAM system was undertaken at Equinor's Troll B field in the Norwegian sector of the North Sea. This location was chosen because it offers a representative offshore production environment with established subsea infrastructure and accessibility for controlled release testing (Metas AS, 2020). The WAAM unit was installed in a dedicated bracket on the Z2 subsea structure near Production Site H, at a depth of approximately three hundred meters, and was connected to the existing TechnipFMC subsea control system to allow integration with the field's operational monitoring framework (Metas AS, 2020).

Controlled leak scenarios were generated using a purpose-built Calibration Frame (Figure 4), deployed independently on the seabed and developed with support from Innovation Norway and Equinor (Metas AS, 2022). The unit can release compressed air to simulate methane or free gas at pressures up to three hundred bars, and 'green' liquid, rapeseed oil, up to two hundred liters to simulate subsea oil releases (Metas AS, 2022). The frame could execute up to ten distinct leakage scenarios, each defined by release type, quantity, and duration. Importantly, the Calibration Frame and WAAM were entirely independent systems, with no data or control connections between them. All release quantities and rates were selected by Equinor personnel and programmed into the Calibration Frame's own control system by use of removable plugs with different resistors installed (Metas AS, 2022). Before each test, control room and offshore personnel were informed so that any alarms raised by WAAM during the test would be recognized as part of planned activities and not acted upon.

Figure 4—Image of Calibration Frame. Source: Imenco Future Technologies

Two major campaigns were conducted. The first, in November 2019, was a straightforward validation to confirm that WAAM met Equinor's minimum performance criteria. The primary goal was to alarm on gas releases of < 10 L/min at a range of 500 meters, while the stretch goal was to alarm on < 100 L/min at 1,000 meters (Metas AS, 2020). Controlled gas releases were conducted at distances from 350 to 950 meters to verify these thresholds (Metas AS, 2022). The second campaign, in 2021, aimed to quantify the actual alarm limits at various ranges. Gas release rates in this phase ranged from 5 L/min, the lowest available calibrated rate, up to 100 L/min, which was the maximum achievable with the available gas flow rate sensor (Metas AS, 2022). Oil releases were fixed at 4 L/min, a constraint of the positive displacement pump used. In both campaigns, multiple release distances (Figure 5) were selected to map alarming performance across WAAM's effective range (Metas AS, 2022). Between campaigns, the system remained deployed to assess long-term reliability and performance under seasonal variations in thermocline and biomass density (Metas AS, 2022).

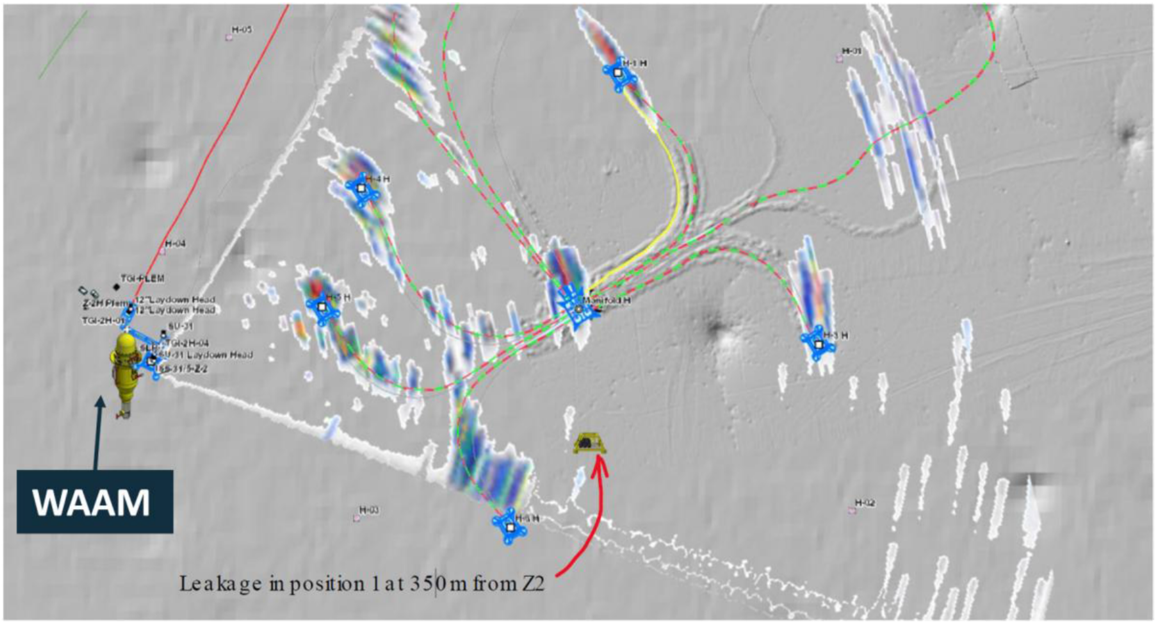

Figure 5—Site Validation Testing of WAAM at Troll B field. Source: Imenco Future Technologies

The WAAM scanning regime for these trials was configured according to the geometry of the field (Metas AS, 2022). A typical full-volume scan consisted of a 91-degree horizontal sweep with six vertical tilt steps and 3.5 degrees of horizontal ping overlap, producing around 156 pings per complete scan. In this configuration, scan cycles took between 280 and 320 seconds to complete (Metas AS, 2022). The scanning sonar operated in a broadband range of 50 to 90 kHz. The scanning sonar employed a single transducer. In other deployments, WAAM can be equipped with a suite of compatible wideband scientific echosounders to match detection requirements (Metas AS, 2022).

All sonar data were processed locally through WAAM's onboard software pipelines. Initial classification separated detected targets into man-made objects, biologics, or leak/seep events (Metas AS, 2022). For targets identified as leaks, a secondary analysis was performed to determine the source location, estimate the volumetric release rate, assess plume stability and growth, and infer fluid type based on acoustic spectral and spatial characteristics (Metas AS, 2022).

Performance was assessed by establishing the lowest flow rate that could be detected consistently across multiple scans, the lowest rate meeting alarm criteria (including persistence and classification confidence), the false alarm rate during both test campaigns and extended deployments, and the accuracy of the system's localization of the leak source relative to the known Calibration Frame position (Metas AS, 2022).

Validation and Field Testing

Validation testing was conducted using the Calibration Frame described in the Methodology section to generate controlled leak scenarios (Metas AS, 2022). The November 2019 campaign confirmed that WAAM met both Equinor's acceptance criteria (Metas AS, 2020). It successfully alarmed on gas releases of < 10 L/min at 500 meters and achieved the stretch goal of alarming on < 100 L/min at 1,000 meters. These results were achieved in conditions of light swell (1–2 meters), water temperatures of 6–10°C, and currents below 0.5 m/s (Metas AS, 2022).

The 2021 campaign provided a more detailed performance map. Gas releases as low as 5 L/min, the minimum measurable flow with the installed sensor triggered alarms at shorter ranges, while the upper limit of 100 L/min, determined by the available sensor control capacity, was also reliably detected (Figure 6). Oil releases, fixed at 4 L/min due to the positive displacement pump, were detected but were not at alarm levels due to the use of rapeseed oil. This oil has an acoustic reflection coefficient that is 1/10th that of liquid hydrocarbon oil, so the 4 liters released was assessed as <0.5 L/min (Metas AS, 2022). Across all test points, no false alarms were recorded (Metas AS, 2022). The extended deployment of over a year between campaigns confirmed that seasonal environmental changes, including shifts in thermocline depth and variations in biomass density, did not degrade detection performance (Metas AS, 2022).

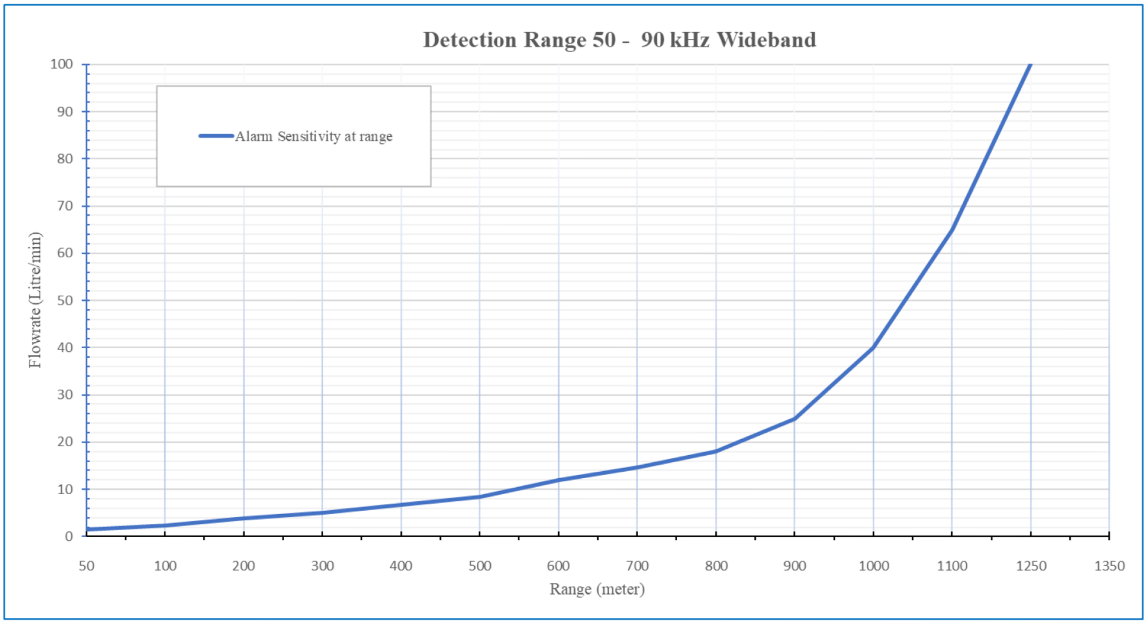

Figure 6—WAAM Alarm Performance in Liters per minute release rate for Gas using Transducer frequency 50-90 kHz (at 300 m water depth) Source: Imenco Future Technologies

System qualification was undertaken in parallel with field validation (Metas AS, 2022). WAAM underwent API 17F qualification, including shock, vibration, and extended lifetime testing (Nemko, 2022), and followed the DNV RP-A203 process for technology readiness, which included a full Failure Modes, Effects, and Criticality Analysis (FMECA) in collaboration with DNV and Equinor personnel (DNV, 2011). Laboratory and harbor trials were also conducted to verify backscatter calibration, avoid near-field saturation, and confirm beam shape under both static and current-affected conditions.

Following these trials, WAAM was accepted as a permanent part of Troll B's subsea monitoring architecture (Metas AS, 2022). It has remained deployed for over five years, with no unplanned downtime and no detections outside of planned calibration events, demonstrating both its long-term operational reliability and its suitability for integration into existing subsea control systems (Metas AS, 2022).

Supporting Research and Independent Validation

The sensor physics and target discrimination capabilities of WAAM are supported by an established body of published research on broadband active acoustics, gas plume characterization, and leak detection in subsea environments.

Previous studies have demonstrated that broadband split-beam echo sounders can reliably detect and characterize bubble plumes across a range of gas compositions, bubble sizes, and flow rates. Loranger (2019) investigated how hydrocarbon droplets of different shapes scatter sound across broadband frequencies. The results confirmed that oil droplets produce distinctive acoustic signatures, reinforcing the basis for WAAM's frequency-domain classification of hydrocarbon versus gas releases. Kubilius (2015) demonstrated that gas bubble plumes and biological scatterers produce distinct spectral and angular acoustic responses. These findings provided reference data that were later incorporated into WAAM's classification logic, particularly in distinguishing leaks from fish or other marine life. Pedersen & Kubilius (2019) showed how split-beam echo sounders could characterize artificial CO_2 plumes in controlled trials, providing quantitative observations of plume rise, structure, and acoustic response. This work validated the feasibility of broadband sonar for detecting small releases, supporting its later use in subsea monitoring.

The ACT4Storage project, conducted independently after the development of WAAM, provided additional validation of acoustic sensing for subsea CO_2 monitoring. Using the same Simrad ES70 and ES333 broadband echosounders commonly deployed in WAAM, the project demonstrated reliable detection of low-rate plumes and characterized their vertical extent, scattering properties, and temporal dynamics. Although WAAM was not part of ACT4Storage, the results closely aligned with WAAM's own field observations, offering third-party confirmation of detection thresholds and plume behavior under offshore conditions (ACT4Storage Project, 2022).

This correlation between WAAM's in-field performance and ACT4Storage's independent findings reinforces the reliability of its detection algorithms, particularly in low-flow scenarios. Furthermore, ACT4Storage's reported detection thresholds are well below the typical alarming limits set for offshore operational deployments (WAAM alarming thresholds are field-configurable to balance sensitivity against false alarm risk) (Metas AS, 2022).

The alignment of laboratory, field, and third-party validation datasets enhances confidence in WAAM's classification performance under diverse operational and environmental conditions. This cross-validation approach—integrating internal test results with peer-reviewed studies and independent trials—meets best practice expectations for offshore monitoring technology qualification (DNV, 2011; Offshore Norge, 2023).

Discussion

The WAAM system has demonstrated capabilities, wide-area scanning, low detection thresholds, and autonomous classification, positioning it as a unique and valuable component of offshore integrity management strategies. Unlike surface monitoring tools such as satellite SAR, optical systems, or radar-based OSD, WAAM provides persistent subsea surveillance, detecting leaks before hydrocarbons or CO_2 reach the surface (Offshore Norge, 2023; ScanEx, 2018).

Integration with Other Monitoring Strategies

WAAM is most effective when used as part of a hybrid detection approach. For example, in carbon capture and storage (CCS) projects, WAAM can be deployed alongside chemical sensors for dissolved CO_2 and optical imaging for direct visual confirmation. In pipeline operations, combining WAAM with distributed acoustic sensing (DAS) or subsea pressure monitoring enables rapid source localization and severity assessment while reducing false alarm rates (Prakoso et al., 2022).

Barrier Management Context

Norwegian petroleum regulations define barriers not only as physical components but as *functional systems* that must be identified, maintained, and verified to meet specified performance standards (Norwegian Petroleum Safety Authority, 2024b). Within this philosophy, WAAM acts as a functional barrier element:

Detection: Early identification of small leaks (down to 5 L/min gas release) before escalation.
Classification: Automated differentiation between leaks, biological activity, and man-made objects.
Mapping: Real-time determination of leak source location, plume trajectory, and likely fluid type.

This capability directly addresses Section 57 of the Activities Regulations (Norwegian Petroleum Safety Authority, 2024a), which requires detection systems to operate independently of light, visibility, and weather, and to rapidly map source and spread.

Operational Scenarios

WAAM offers advantages in fields where subsea infrastructure is distributed or located in areas with limited physical access. Applicable scenarios include:

Tie-in spools and manifolds where multiple leak sources may exist in proximity.
CCS injection wells requiring monitoring for both hydrocarbon and CO_2 leaks.
Pipeline End Terminations (PLETs) and Manifolds (PLEMs) where localized coverage is sufficient to mitigate high-risk points.
Subsea trees and wellheads where permanent surveillance can replace or reduce costly inspection campaigns.

The system is also deployable in time-limited campaigns, such as during commissioning, integrity verification after intervention, or targeted inspections prompted by anomalies in other monitoring data. In such cases, WAAM's redeployable lander configuration enables rapid relocation between pre-installed landing sites.

Detection vs. Alarming Limits

A key distinction in interpreting WAAM's performance lies between detection limits and alarming limits. While the system can detect plumes at rates well below 1 L/min—supported by ACT4Storage findings for similar sensors (Loranger, 2022)—operational alarming thresholds are typically set higher to ensure persistence, classification confidence, and low false alarm rates. These thresholds are configurable, enabling optimization for site-specific risk tolerance and environmental conditions.

Limitations and Scalability

WAAM's practical range (typically 1–2 km radius) means that monitoring long pipelines may require multiple units or rotational deployment strategies. However, modularity, low power requirements, and compatibility with existing subsea infrastructure reduce logistical barriers to scaling coverage. Fixed landing points or receptacles can further streamline redeployment.

Conclusion

Field validation at Equinor's Troll B field, supported by controlled leak simulations using an independent calibration frame, confirmed that the Wide Area Active Monitoring (WAAM) system meets and exceeds defined operational performance targets for subsea leak detection. Gas leaks as low as 5 L/min and oil releases of 4 L/min were reliably detected at operationally relevant ranges, with no false alarms recorded during extended deployments. These results, coupled with over five years of continuous subsea operation without unplanned downtime, demonstrate the system's robustness and suitability for integration into existing subsea monitoring architectures.

WAAM's design, a compact, seabed-mounted broadband sonar with programmable scanning and onboard machine learning classification, provides persistent, wide-area coverage independent of light, visibility, or weather. Its ability to classify and map both hydrocarbon and CO_2 plumes in real time addresses the requirements of Norwegian Activities Regulation §57 and supports a barrier management approach in which leak detection forms part of the functional safety envelope.

The system's operational flexibility allows deployment around distributed subsea assets, including manifolds, tie-in spools, PLETS, PLEMs, subsea trees, and CCS injection wells. It is equally applicable for temporary monitoring during commissioning, post-intervention verification, or targeted inspections prompted by anomalies in other surveillance data. The distinction between detection limits and alarming thresholds, configurable to suit site-specific risk tolerance, ensures that operational alarms are both sensitive and reliable.

WAAM's field-proven performance aligns with independent research on acoustic plume detection, such as the ACT4Storage project and published studies on broadband acoustic scattering. This independent alignment strengthens confidence in the system's detection physics and classification methodology.

As offshore industries increasingly adopt real-time monitoring and digital twin integration, WAAM offers a proven and scalable platform for condition-based integrity management, environmental stewardship, and regulatory compliance. Future work may also investigate the adaptation of WAAM's volumetric scanning and automated classification capabilities to other subsea monitoring domains beyond leak detection, leveraging the same architecture to address emerging offshore operational needs.

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References

ACT4Storage Project, 2022. ACT4Storage: Acoustic and Chemical Technologies for Environmental Monitoring of Geological Carbon Storage. Presented by S. Loranger, *ACT4Storage Project Summary Session*, Bergen, Norway, 10 December 2019. Available at: https://vimeo.com/375740088 (accessed August 2025).

API, 2017. Specification 17F: Subsea Production Control Systems. American Petroleum Institute, Washington, D.C.

DNV, 2011. Recommended Practice DNV-RP-F302: Selection and Use of Subsea Leak Detection Systems. Det Norske Veritas, Høvik, Norway.

DNV, 2021. DNV-RP-A203: Technology Qualification. DNV, Høvik, Norway.

Fingas, M., and Brown, C., 2018. "Oil Spill Remote Sensing: A Review." *Marine Pollution Bulletin*, **134**: 1–17. https://doi.org/10.1016/j.marpolbul.2017.09.030

Imenco Future Technologies, 2024. WAAM Product Data Sheet. Internal document, Imenco Future Technologies AS, Bergen, Norway. (Available upon request.)

Imenco Future Technologies, 2023. WAAM Software Release Notes. Internal document, Imenco Future Technologies AS, Bergen, Norway. (Available upon request.)

Jha, M. N., Levy, J., Gao, Y., 2008. "Advances in Remote Sensing for Oil Spill Disaster Management: State-of-the-Art Sensors Technology for Oil Spill Surveillance." *Sensors*, **8**(1): 236–255. https://doi.org/10.3390/s8010236

Johannessen, O. M., Kudryavtsev, V., Akimov, D., et al, 2006. "On Radar Imaging of Current Features: 2. Mesoscale Eddy and Current Front Detection." *Journal of Geophysical Research*, **111**, C09016. https://doi.org/10.1029/2005JC003252

Kongsberg Discovery, 2024. Ocean Science Transducers – Product Catalogue. Kongsberg Gruppen ASA, Kongsberg, Norway. Available at: https://www.kongsberg.com/discovery/ocean-science/ocean-science-transducers/

Kubilius, R., 2015. Acoustic Discrimination of Biological Targets and Gas Plumes in the Ocean. PhD thesis, University of Bergen, Norway. Available at: https://bora.uib.no/bora-xmlui/handle/1956/11040

Loranger, S., 2019. "Broadband Acoustic Scattering from Oblate Hydrocarbon Droplets." *Journal of the Acoustical Society of America*, **146**(5): 3364. https://doi.org/10.1121/1.5137740

Metas AS, 2020. Troll B WAAM Initial Field Trial Results. Internal report, Metas AS, Bergen, Norway. (Available upon request.)

Metas AS, 2022. Metas Subsea Leak Detection Systems NOROG Oil and Gas Subsea Templates. Presentation to NOROG (now Offshore Norge), 24 March 2022, Bergen, Norway.

Norwegian Petroleum Safety Authority, 2024. Activities Regulations §57: Detection and Mapping of Acute Pollution. Available at: https://www.havtil.no/en/regulations/all-acts/the-activities-regulations3/X/57/

Norwegian Petroleum Safety Authority, 2024b. Management Regulations §5: Barriers. Available at: https://www.havtil.no/en/regulations/all-acts/the-management-regulations3/X/5/

Offshore Norge, 2023. Guideline 100: Recommended Guidelines for Detection of Acute Discharges (Rev. 04). Offshore Norge, Stavanger, Norway. Available at: https://www.offshorenorge.no/contentassets/4bdac57a10e648e797649df5b7471eba/100---recommended-guidelines-for-detection-of-acute-discharges-rev.-04.pdf

OGP, 2013. Report 477: Environmental Performance Indicators – 2012 Data. International Association of Oil & Gas Producers, London, UK.

Pedersen, G., and Kubilius, R., 2019. "Acoustic Characterization of an Artificial CO_2 Seep Using Split-Beam Echo Sounders." Proceedings of the UACE2019 Conference, Hersonissos, Crete, Greece.

Prakoso, T., et al, 2022. "A Review of Acoustic Leak Detection Methods in Subsea Applications." *Journal of Marine Science and Engineering*, **10**(4): 514. https://doi.org/10.3390/jmse10040514

Reddy, C. M., Arey, J. S., Seewald, J. S., et al, 2012. "Composition and Fate of Gas and Oil Released to the Water Column During the Deepwater Horizon Oil Spill." *Proceedings of the National Academy of Sciences*, **109**(50): 20229–20234. https://doi.org/10.1073/pnas.1101242108

ScanEx, 2018. Satellite Monitoring of Oil Pollution in the Persian Gulf. ScanEx Research and Development Center, Moscow, Russia. Available at: https://www.scanex.ru/en/company/news/satellite-monitoring-of-oil-pollution-in-the-persian-gulf/